Summary for phase 1

March 4, 2026

Baseline parameters. Unless otherwise specified, we adopt the following fiducial parameters in the model:

$$A_{\rm cool} = 1.0 \tag{1}$$
$$t_{\rm dep,0} = 1.0 \tag{2}$$
$$t_{\rm dep,exp} = 0.0 \tag{3}$$
$$\eta_{M,0} = 1.0 \tag{4}$$
$$\eta_{M,\rm exp} = 1.0 \tag{5}$$
$$\eta_{E,0} = 1.0 \tag{6}$$
$$\eta_{E,\rm exp} = 0.0 \tag{7}$$
$$A_{\rm heat} = 1.0 \tag{8}$$
$$A_{\rm cool,CGM} = 1.0 \tag{9}$$

The simulation is evolved to a final time $t_{\rm end} = 30$ with $n_{\rm steps} = 1000$ integration steps.

1 Effect of the Star Formation Nonlinearity

In our model, the star formation rate (SFR) is determined by the depletion time prescription

$$t_{\rm dep} = t_{\rm dep,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{t_{\rm dep,exp}}, \tag{10}$$

where $M_{\rm ISM}$ is the mass of the interstellar medium (ISM), $t_{\rm dep,0}$ is the normalization of the depletion time, and $t_{\rm dep,exp}$ controls the dependence of the depletion time on the ISM mass.

The star formation rate is then

$$\dot{M}_\star = \frac{M_{\rm ISM}}{t_{\rm dep}}. \tag{11}$$

Substituting the depletion time expression gives

$$\dot{M}_\star = \frac{M_{\rm ISM}}{t_{\rm dep,0}} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{-t_{\rm dep,exp}} \propto M_{\rm ISM}^{1-t_{\rm dep,exp}}. \tag{12}$$

In this experiment we keep all parameters fixed except $t_{\rm dep,exp}$, and explore

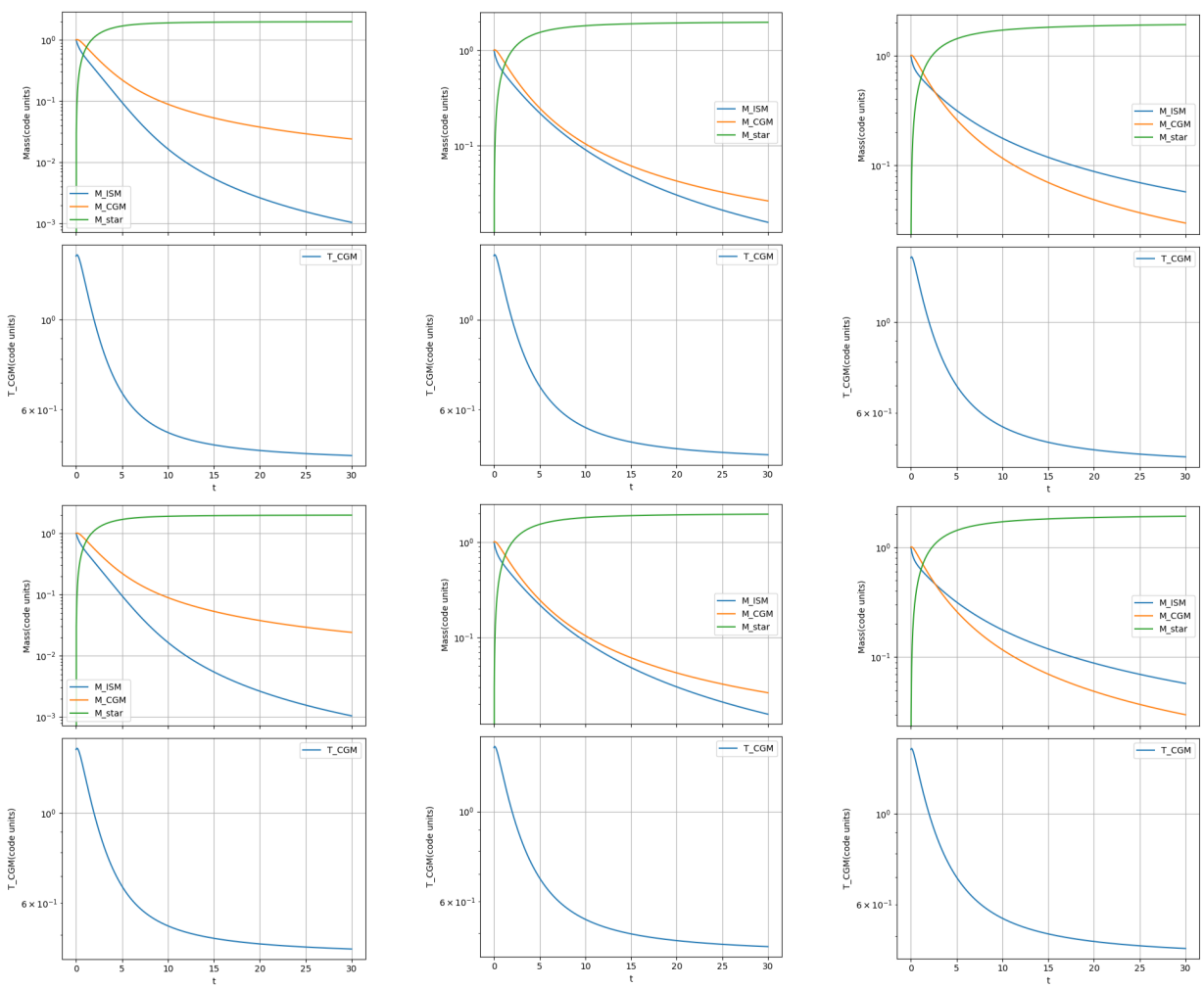
$$t_{\rm dep,exp} \in \{0,\ -0.5,\ -1\}. \tag{13}$$

These values correspond to different scaling relations:

$$t_{\rm dep,exp} = 0 \quad \Rightarrow \quad \dot{M}_\star \propto M_{\rm ISM}, \tag{14}$$

$$t_{\rm dep,exp} = -0.5 \quad \Rightarrow \quad \dot{M}_\star \propto M_{\rm ISM}^{1.5}, \tag{15}$$

$$t_{\rm dep,exp} = -1.0 \quad \Rightarrow \quad \dot{M}_\star \propto M_{\rm ISM}^{2}. \tag{16}$$



(a) $t_{\rm dep,exp} = 0$ (b) $t_{\rm dep,exp} = -0.5$ (c) $t_{\rm dep,exp} = -1$

Figure 1: Evolution of $M_{\rm ISM}$, $M_{\rm CGM}$, $M_\star$, and $T_{\rm CGM}$ for different values of the star formation nonlinearity parameter $t_{\rm dep,exp}$. More negative values correspond to stronger nonlinear dependence of the star formation rate on the ISM mass.

From Fig. 1 we see that varying the depletion-time exponent $t_{\rm dep,exp}$ changes the nonlinearity of the star formation law and significantly affects how rapidly the ISM is consumed.

For $t_{\rm dep,exp} = 0$, the depletion time is constant and the star formation rate scales linearly with the ISM mass, $\dot{M}_\star \propto M_{\rm ISM}$. In this case the ISM mass declines gradually over time while the stellar mass grows smoothly as gas is converted into stars.

When $t_{\rm dep,exp} = -0.5$, the star formation rate becomes more sensitive to the ISM mass, corresponding to $\dot{M}_\star \propto M_{\rm ISM}^{1.5}$. As a result, star formation is enhanced at early times when the ISM reservoir is large. This leads to a faster depletion of the ISM and a more rapid buildup of stellar mass.

For $t_{\rm dep,exp} = -1$, the dependence becomes even stronger, $\dot{M}_\star \propto M_{\rm ISM}^{2}$. The ISM is consumed most rapidly in this case, producing a very steep decline of $M_{\rm ISM}$ at early times and a rapid growth of $M_\star$. After the gas reservoir is largely depleted, the system approaches a quasi-steady state.

In all three cases the CGM temperature initially decreases and then gradually approaches an equilibrium value. The temperature evolution is only weakly affected by the change in $t_{\rm dep,exp}$ because this parameter mainly controls the efficiency of star formation rather than the heating and cooling balance in the CGM.

2 Effect of Wind Mass Loading

We model the galactic wind mass outflow rate as proportional to the star formation rate:

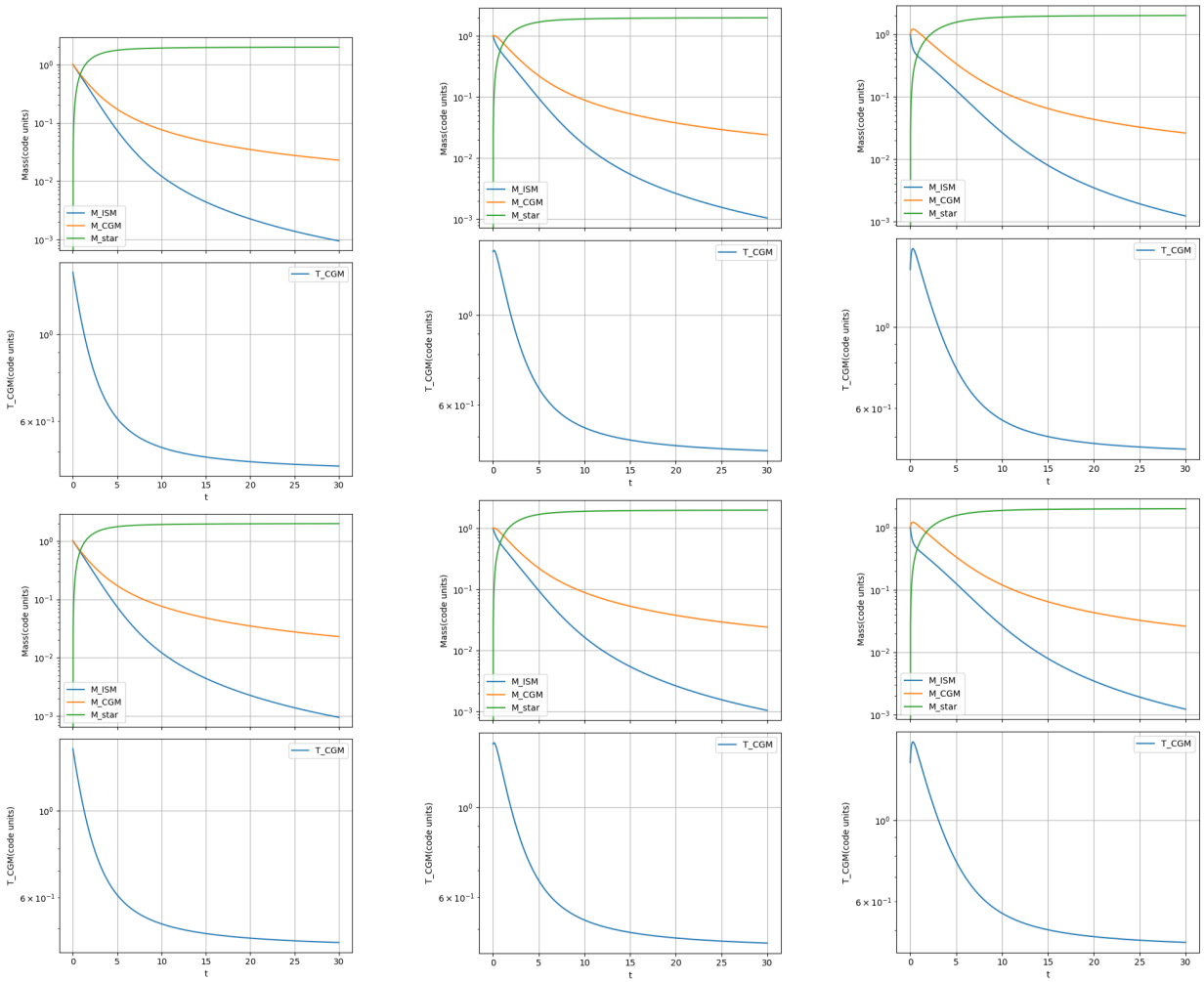
$$\dot{M}_{\rm wind} = \eta_M \, \dot{M}_\star. \tag{17}$$

In our implementation, the mass loading factor depends on the ISM mass as

$$\eta_M = \eta_{M,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{\eta_{M,exp}}, \tag{18}$$

where $\eta_{M,0}$ sets the overall normalization. In this experiment we fix all parameters except $\eta_{M,0}$ and explore

$$\eta_{M,0} \in \{0.3,\ 1.0,\ 3.0\}. \tag{19}$$



(a) $\eta_{M,0} = 0.3$ (b) $\eta_{M,0} = 1.0$ (c) $\eta_{M,0} = 3.0$

Figure 2: Time evolution of $M_{\rm ISM}$, $M_{\rm CGM}$, $M_\star$, and $T_{\rm CGM}$ for different values of the wind mass-loading normalization $\eta_{M,0}$.

From Fig. 2 we see that varying the wind mass-loading normalization $\eta_{M,0}$ changes the strength of galactic winds and therefore affects how efficiently gas is removed from the ISM.

For smaller values of $\eta_{M,0}$ (e.g., $\eta_{M,0} = 0.3$), the winds are relatively weak and only a modest amount of gas is expelled from the ISM. In this case the ISM mass decreases primarily due to star formation, while the CGM mass grows more gradually as only a small fraction of gas is transported into the halo.

When $\eta_{M,0} = 1$, the wind strength is moderate and the outflow rate becomes comparable to the star formation rate. As a result, gas is removed from the ISM more efficiently and the CGM reservoir grows more rapidly.

For larger values of $\eta_{M,0}$ (e.g., $\eta_{M,0} = 3$), the galactic winds become stronger and a larger fraction of the ISM gas is transferred into the CGM. This leads to a slightly faster decline of $M_{\rm ISM}$ and a correspondingly larger buildup of $M_{\rm CGM}$.

Despite these differences in gas redistribution, the stellar mass evolution remains broadly similar across the three cases. In addition, the CGM temperature evolution is nearly un-

changed, gradually approaching a quasi-steady value as heating and cooling processes balance each other.

2.1 Effect of the Wind Mass-Loading Exponent

Galactic winds remove gas from the interstellar medium (ISM) at a rate proportional to the star formation rate,

$$\dot{M}_{\rm wind} = \eta_M \, \dot{M}_\star \tag{20}$$

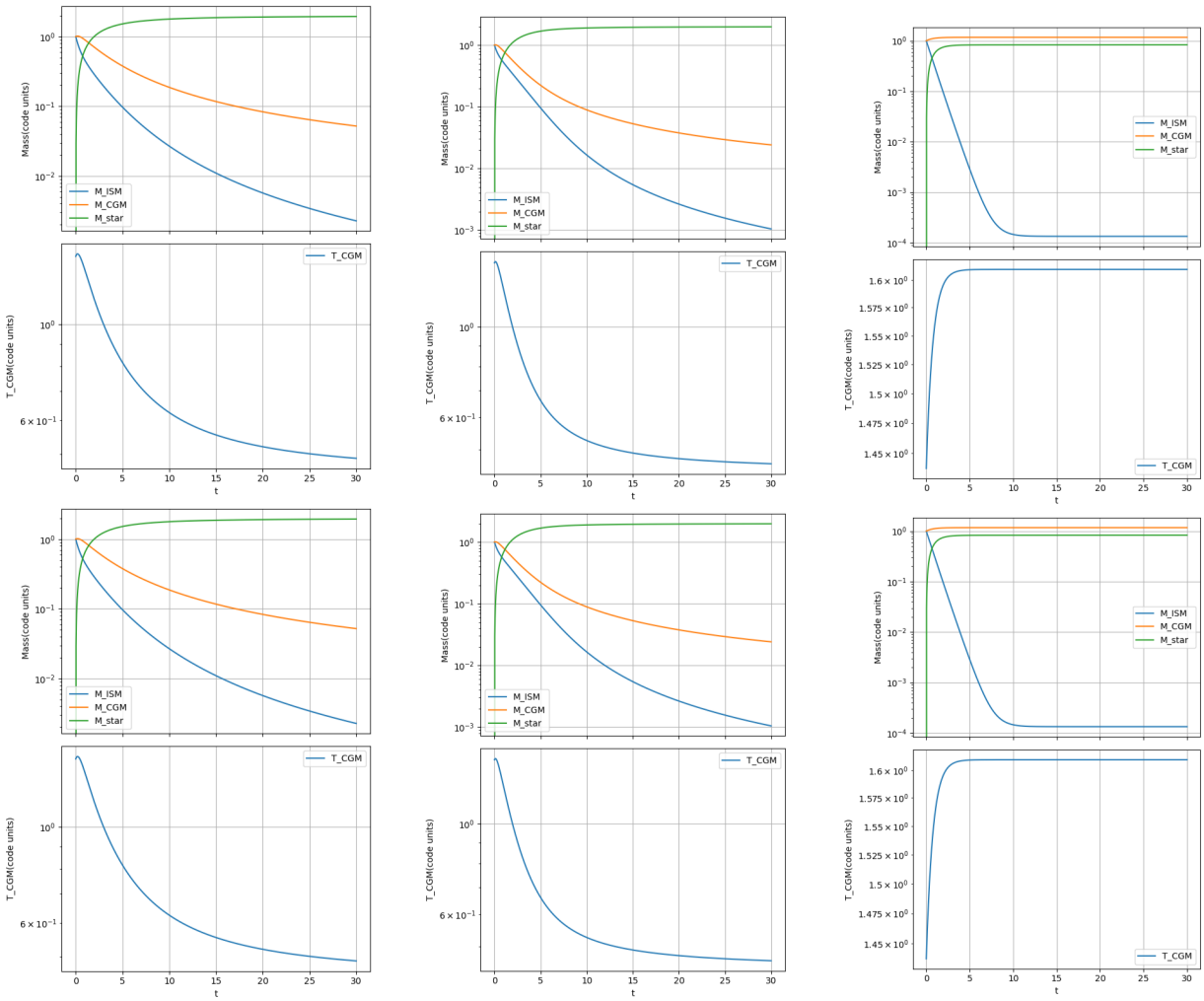
The mass-loading factor is parameterized as

$$\eta_M = \eta_{M,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{\eta_{M,\rm exp}} \tag{21}$$

where $\eta_{M,0}$ sets the normalization and $\eta_{M,\rm exp}$ controls the dependence of the wind strength on the ISM mass.

In this experiment we keep all other parameters fixed and explore

$$\eta_{M,\rm exp} \in \{-1,\, 0,\, 1\}. \tag{22}$$



(a) $\eta_{M,\mathrm{exp}} = 0$ (b) $\eta_{M,\mathrm{exp}} = 1$ (c) $\eta_{M,\mathrm{exp}} = -1$

Figure 3: Time evolution of M_{ISM}, M_{CGM}, $M_\star$, and T_{CGM} for different values of the wind mass-loading exponent $\eta_{M,\mathrm{exp}}$. Panels (a)–(c) correspond to $\eta_{M,\mathrm{exp}} = 0$, 1, and -1, respectively.

From Fig. 3 we see that varying the wind mass-loading exponent $\eta_{M,\mathrm{exp}}$ changes how strongly the wind efficiency depends on the ISM mass.

For $\eta_{M,\mathrm{exp}} = 0$, the mass-loading factor remains constant and the wind strength does not depend on the ISM mass. In this case the ISM mass declines smoothly as gas is converted into stars and partially expelled into the CGM, while the stellar mass grows steadily.

When $\eta_{M,\mathrm{exp}} = 1$, the wind becomes stronger when the ISM mass is large. As a result, gas is removed more efficiently at early times when the ISM reservoir is abundant. This leads to a slightly faster redistribution of gas into the CGM and a somewhat more rapid decline of M_{ISM} at early times.

For $\eta_{M,\mathrm{exp}} = -1$, the wind efficiency becomes larger when the ISM mass is small. In this case the winds remain relatively weak at early times but become more effective after the ISM has been partially depleted. This tends to maintain a lower ISM reservoir at late times.

Despite these differences in wind scaling, the overall stellar mass growth remains similar

across the three cases. The CGM temperature evolution is also largely unchanged, gradually approaching an equilibrium value as heating and cooling processes balance.

3 Effect of the Wind Heating Normalization

We model wind heating of the CGM as

$$\dot{T}_{\rm CGM}\Big|_{\rm wind} = \eta_E\, \dot{M}_{\rm wind}, \tag{23}$$

where the wind mass outflow rate is

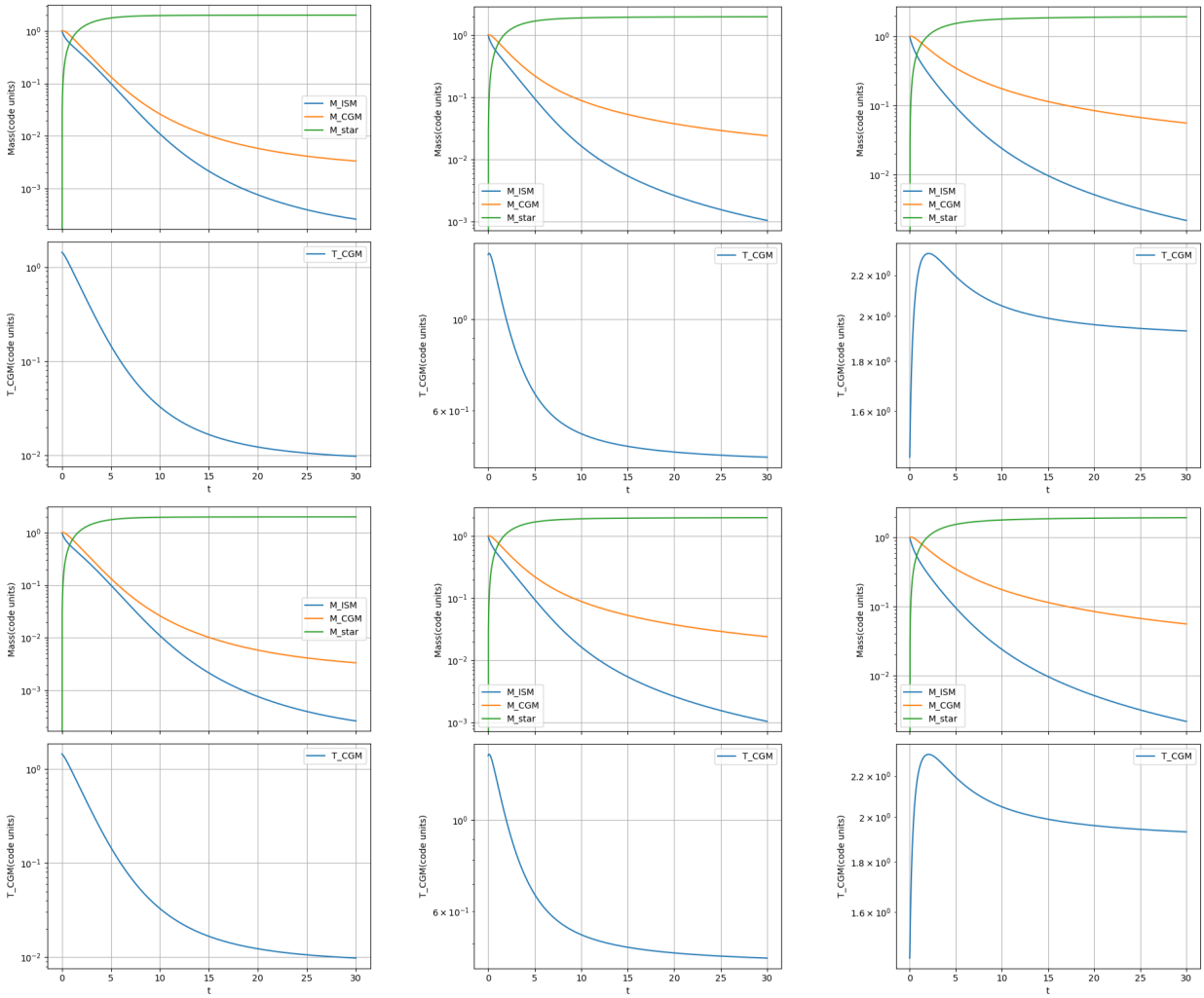
$$\dot{M}_{\rm wind} = \eta_M\, \dot{M}_\star, \qquad \eta_M = \eta_{M,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{\eta_{M,\rm exp}}. \tag{24}$$

The energy-loading factor is parameterized as

$$\eta_E = \eta_{E,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{\eta_{E,\rm exp}}. \tag{25}$$

In this experiment we fix all parameters except $\eta_{E,0}$, set $\eta_{E,\rm exp} = 0$, and explore

$$\eta_{E,0} \in \{0.5,\ 1,\ 3\}. \tag{26}$$



(a) $\eta_{E,0} = 0.5$ (b) $\eta_{E,0} = 1$ (c) $\eta_{E,0} = 3$

Figure 4: Time evolution of $M_{\rm ISM}$, $M_{\rm CGM}$, $M_\star$, and $T_{\rm CGM}$ for different values of the wind-heating normalization $\eta_{E,0}$, with all other parameters fixed and $\eta_{E,\rm exp} = 0$.

Increasing $\eta_{E,0}$ strengthens wind heating, leading to systematically higher $T_{\rm CGM}$ and a more prominent early-time temperature peak. In contrast, the mass evolution ($M_{\rm ISM}$, $M_{\rm CGM}$, and $M_\star$) is only weakly affected, indicating that $\eta_{E,0}$ primarily regulates the thermal state of the CGM rather than the mass flow rates. For sufficiently small $\eta_{E,0}$, the reduced heating can allow strong cooling to drive $T_{\rm CGM}$ to very low values, which may lead to numerical instabilities in cooling-limited inflow prescriptions.

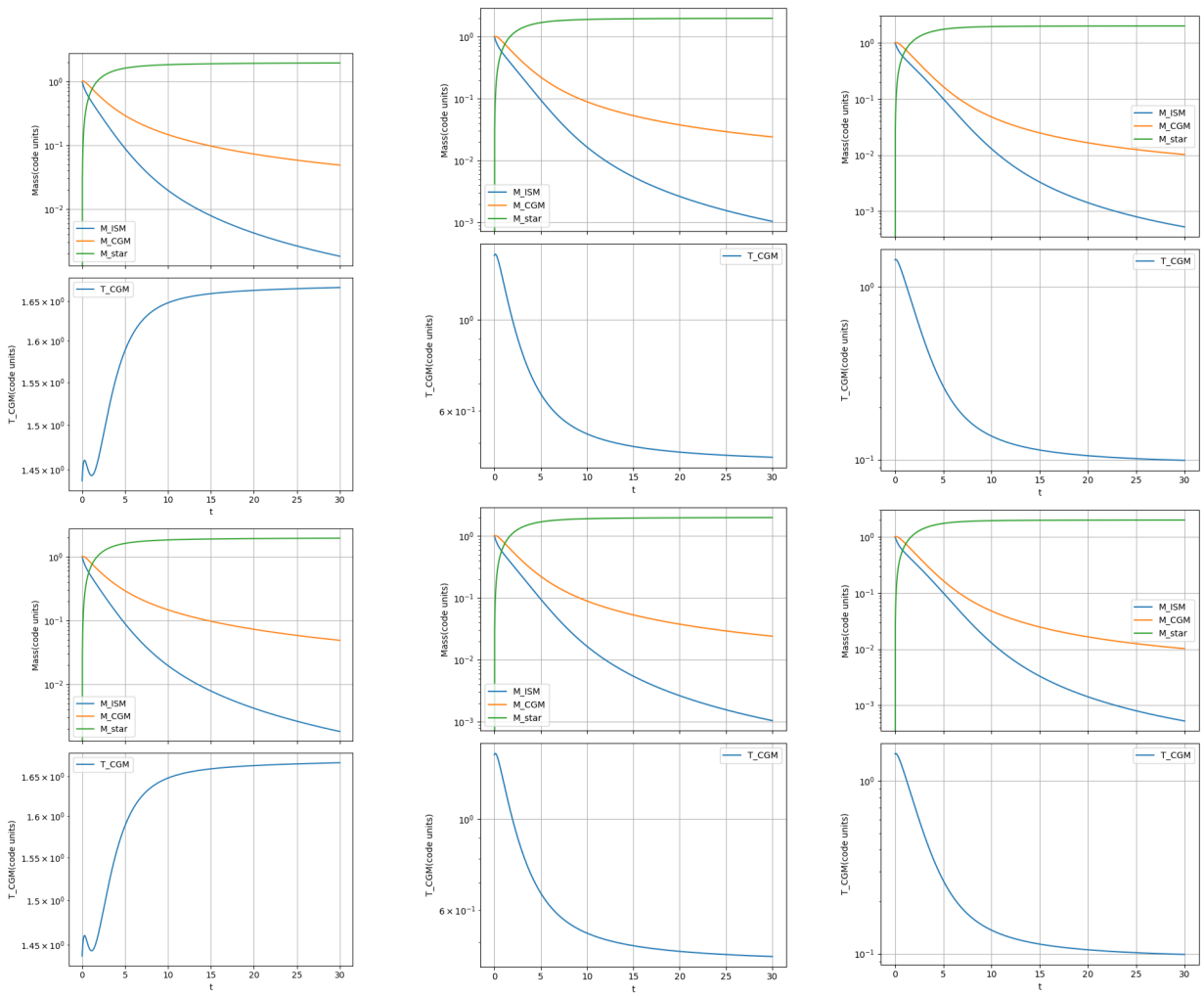
3.1 Effect of the Wind Energy-Loading Exponent

The wind heating efficiency is parameterized as

$$\eta_E = \eta_{E,0} \left(\frac{M_{\rm ISM}}{M_{\rm ISM,ref}} \right)^{\eta_{E,\rm exp}} . \tag{27}$$

In this experiment we keep all other parameters fixed and explore

$$\eta_{E,\mathrm{exp}} \in \{-1, 0, 1\}. \tag{28}$$



(a) $\eta_{E,\mathrm{exp}} = -1$ (b) $\eta_{E,\mathrm{exp}} = 0$ (c) $\eta_{E,\mathrm{exp}} = 1$

Figure 5: Evolution of M_{ISM}, M_{CGM}, $M_\star$, and T_{CGM} for different values of the wind energy-loading exponent $\eta_{E,\mathrm{exp}}$, with $\eta_{E,0} = 1$ and all other parameters fixed.

From Fig. 5 we see that varying the wind energy-loading exponent $\eta_{E,\mathrm{exp}}$ changes how the wind heating efficiency depends on the ISM mass.

For $\eta_{E,\mathrm{exp}} = 0$, the energy-loading factor remains constant, so the wind heating rate scales directly with the wind mass outflow. In this case the CGM temperature gradually decreases and approaches a steady value as heating and cooling balance each other.

When $\eta_{E,\mathrm{exp}} = 1$, the wind heating becomes stronger when the ISM mass is large. This enhances the heating at early times when the gas reservoir is abundant, producing a higher CGM temperature and a more pronounced temperature peak before the system settles toward equilibrium.

For $\eta_{E,\mathrm{exp}} = -1$, the heating efficiency becomes relatively stronger when the ISM mass is small. As a result the CGM temperature increases more gradually and approaches its equilibrium value from below at late times.

Despite these differences in the thermal evolution, the mass evolution of M_{ISM}, M_{CGM}, and $M_\star$ remains very similar across the three cases, indicating that $\eta_{E,\mathrm{exp}}$ primarily affects the thermal state of the CGM rather than the overall mass flow rates in the system.